

JIABIAO SU

C++ SOFTWARE ENGINEER

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PROFILE

C++ software engineer focused on computational geometry, high-performance systems and OpenGL. Led the architecture and 0-to-1 development of industrial 3D-printing software; seeking to apply systems development and performance optimisation skills in FinTech.

TECHNICAL SKILLS

Languages: C++17, C, Python

Graphics and interfaces: OpenGL, Qt, OpenCV, OpenVR

Systems and algorithms: Linux, multithreading, computational geometry, optimisation, TCP, serial communication, STM32

PROFESSIONAL EXPERIENCE

Guangzhou Riton Additive Technology Co., Ltd.

Jun 2024 - Sep 2026

Project Lead / C++ Software Engineer | Guangzhou, China

- Architected and independently developed SLM 3D-printing software from scratch, covering model rendering, geometry, automatic placement and interactive UI.
- Built C++17 computational pipelines with thread pools and parallel tasks; optimised memory access, task granularity and data copying to improve processing efficiency and responsiveness.
- Implemented model geometry calculations, region identification, 2D projection and equidistant sampling, linking 3D model analysis to manufacturing data generation.
- Developed automatic placement and nesting for irregular models, addressing NP-hard combinatorial optimisation while balancing build-platform utilisation and computational cost.
- Created OpenGL rendering modules with rotation, translation, off-screen rendering and multisample anti-aliasing (MSAA), balancing visual quality with real-time performance.

Shenzhen Lian Robotics Co., Ltd.

Sep 2022 - Feb 2024

C++ Software Engineer | Shenzhen, China

- Built Qt/OpenGL tools for optical tracking and robot-arm trajectory capture; used OpenVR for live HTC Tracker pose visualisation and replayed trajectories from files.
- Integrated MCU pressure sensing and serial communication with PCs; developed Android software using TCP to mark selected trajectory segments.
- Developed Linux/Qt control software using robot-arm SDKs for a physiotherapy robot; tested performance, assessed development timelines and advised on equipment procurement.
- Selected sensors, MCUs and VR equipment; coordinated work and tracked project progress with PingCode.

EDUCATION

University of Glasgow

Expected graduation: September 2027

MSc Software Engineering

Guangdong Polytechnic Normal University

Sep 2019 - Jul 2023

Bachelor's degree in Automation

Relevant modules: Advanced C++ Programming, Embedded Systems and Applications, Computing, Communication Technology, Calculus, Linear Algebra.

SELECTED UNIVERSITY PROJECTS

National University Student Smart Car Competition

- Used a USB camera and OpenCV for lane detection on a Linux single-board computer; calculated tracking error and transmitted it to STM32 for PID vehicle control.

AI-based Grocery-Picking Robot

- Trained models with Baidu PaddlePaddle and deployed them to a PC; implemented TCP image exchange and recognition-result handling, with serial commands sent to a lower-level controller.

AWARDS AND LANGUAGES

Second Prize, 17th National University Student Smart Car Competition | Aug 2022

First Prize, 16th National University Student Smart Car Competition | Aug 2021

Provincial award, Electronics Design Competition | 24-25 Apr 2021

English: IELTS 6.5